Week 3, Lecture 1
Quasi-probability distributions

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Notes 6
Bachor & Ralph, 4.3.2
Walls & Milburn, 3.2

Review

What we covered last week…

States

Fock photon number states

$$\{|n\rangle\} = \{|0\rangle, |1\rangle, |2\rangle, \cdots\}$$

Coherent states

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_n \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

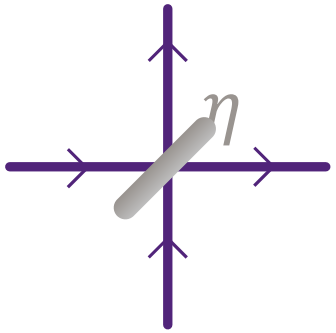
Thermal states

$$\hat{\rho} = \frac{1}{\bar{n}+1} \sum_n \left(\frac{\bar{n}}{\bar{n}+1}\right)^n |n\rangle\langle n|$$

Operations

Beam splitters

$$\hat{a}_1' = \sqrt{\eta}\,\hat{a}_1 - \sqrt{1-\eta}\,\hat{a}_2$$
$$\hat{a}_2' = \sqrt{1-\eta}\,\hat{a}_1 + \sqrt{\eta}\,\hat{a}_2$$

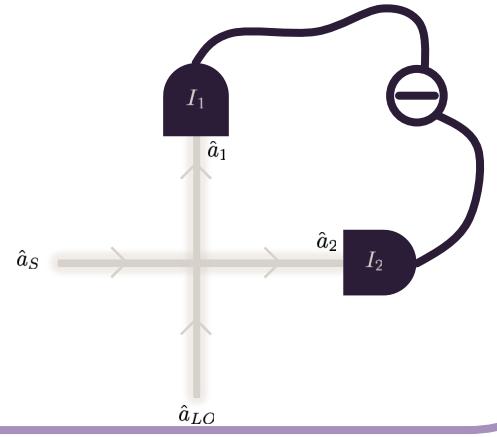


Phase shift

$$\hat{U}(\theta) = \exp(-i\theta\hat{n})$$

Measurement

- Avalanche photodiodes are sensitive enough to detect single photons
- Homodyne detection is used to measure quadratures $\hat{x}$ and $\hat{p}$

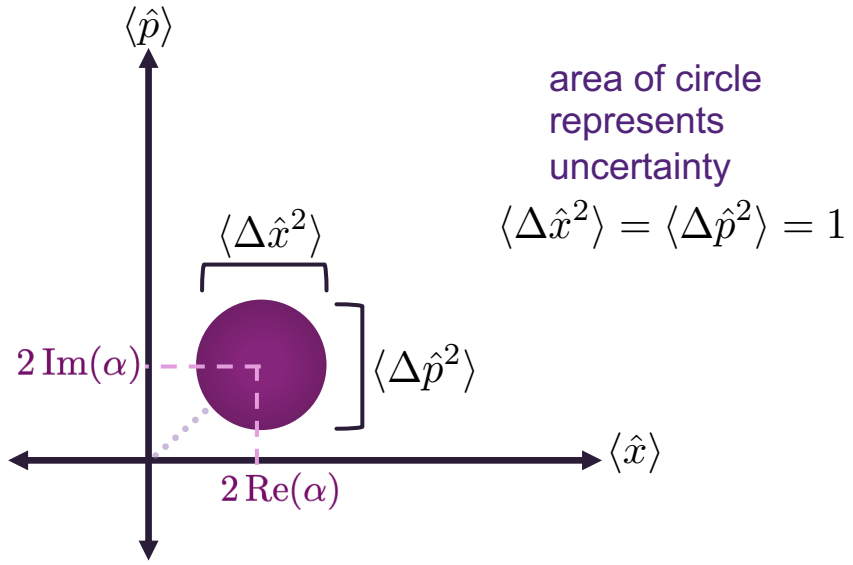


Phase space - review

Quadrature operators can be used to visualise states on a two-dimensional axis

$$\hat{x} = \left(\hat{a}^\dagger + \hat{a}\right)$$
$$\hat{p} = i\left(\hat{a}^\dagger - \hat{a}\right)$$

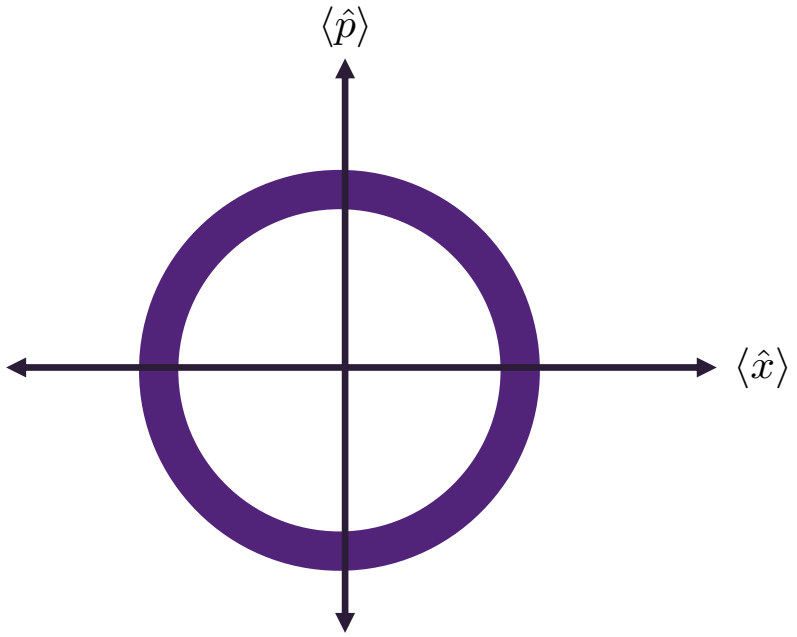
X and P can be measured by homodyne detection



area of circle represents uncertainty

$$\langle\Delta\hat{x}^2\rangle = \langle\Delta\hat{p}^2\rangle = 1$$

Coherent state



Number state

Quasi-probability distributions

Overview

In this lecture, we will look at three different quasi-probability distributions used in quantum optics:

Husimi Q function

Glauber Sudarshan P function

Wigner function

- Useful for visualising quantum states and their evolution
- Each of the Q, P and W functions have their own advantages and disadvantages
- The Q, P and W functions are related to each other
- *Why are they called quasi-probability distributions?*
  - Could be negative or ill-behaved

Q function

Definition

The Q function is defined as the overlap with a coherent state

$$Q(\alpha) = \frac{1}{\pi} |\langle \psi | \alpha \rangle|^2$$ pure state

$$Q(\alpha) = \frac{1}{\pi} \langle \alpha | \hat{\rho} | \alpha \rangle$$ general

gives the probability distribution for finding the coherent state $|\alpha\rangle$ in the state $\hat{\rho}$

The Q function is:

- non-negative
- normalised to unity $\int Q(\alpha) \mathrm{d}^2\alpha = 1$

completeness relation of coherent states

$$\mathbb{1} = \frac{1}{\pi} \int |\alpha\rangle \langle \alpha| \, \mathrm{d}^2\alpha$$

- bounded by $\frac{1}{\pi}$

$$0 \leq Q(\alpha) \leq \frac{1}{\pi}$$

Q function can be thought of as a probability density (P and W functions cannot)

Q function

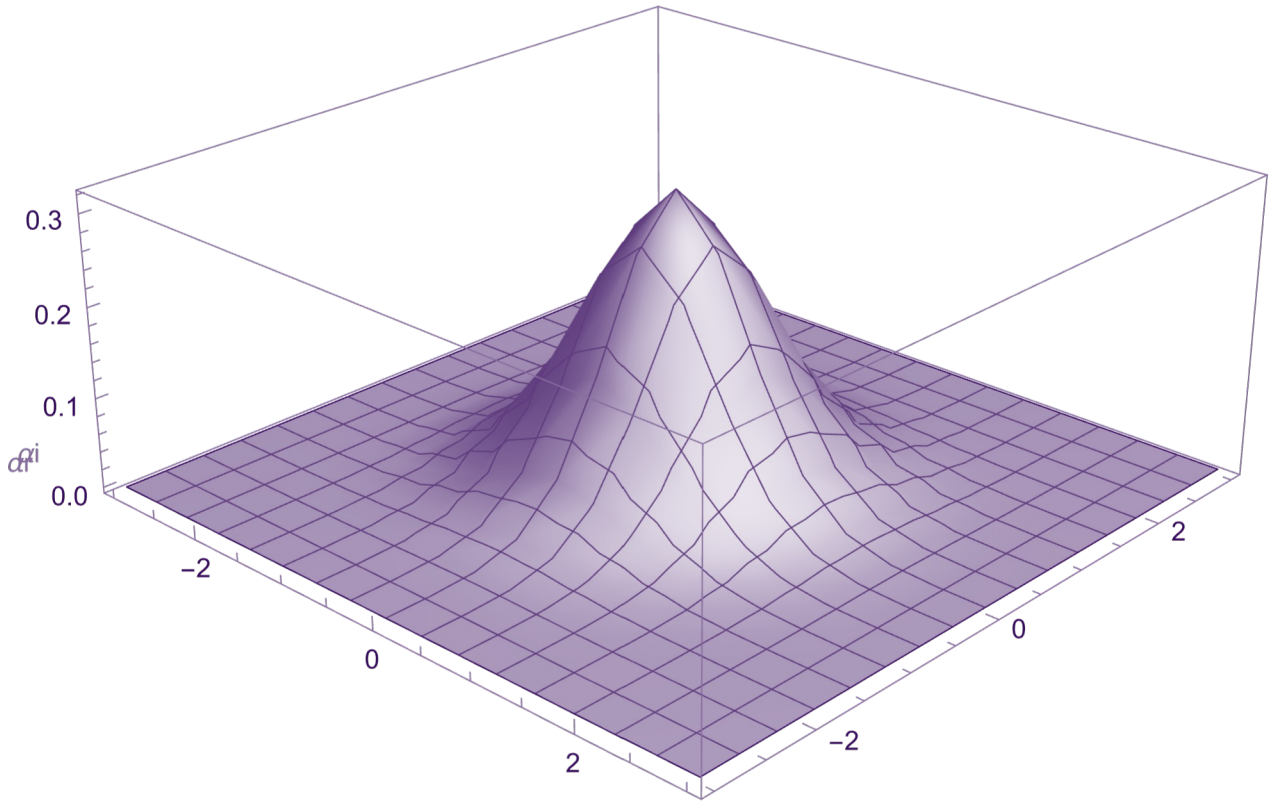
Example: vacuum state

Q function of $|0\rangle$

$$Q(\alpha) = \frac{1}{\pi} |\langle 0|\alpha\rangle|^2$$

$$Q(\alpha) = \frac{1}{\pi} e^{-|\alpha|^2}$$

Gaussian centred at (0,0)



Q function

Example: coherent state

Q function of $|\beta\rangle$

$$Q(\alpha) = \frac{1}{\pi} |\langle\beta|\alpha\rangle|^2$$

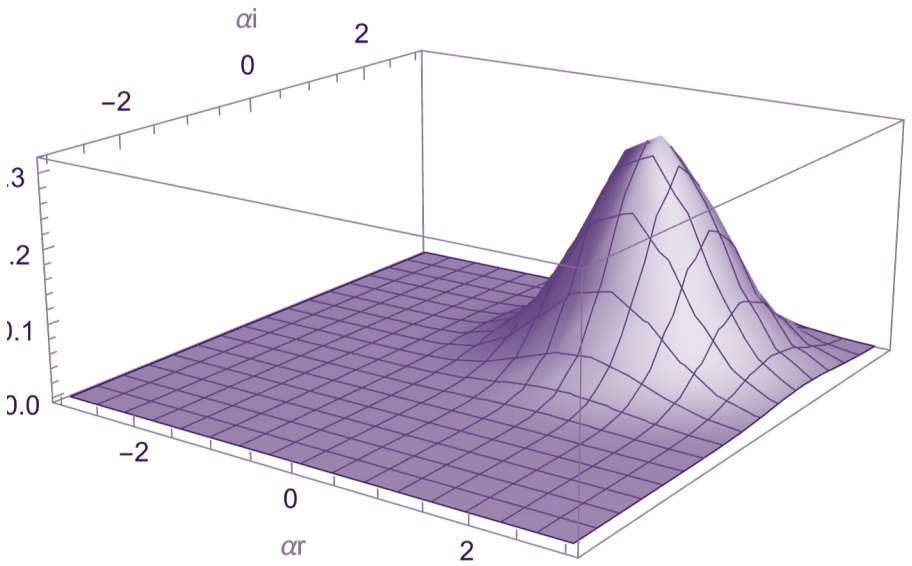
$$Q(\alpha) = \frac{1}{\pi} e^{-|\alpha-\beta|^2}$$

Gaussian centred at $(Re(\beta), Im(\beta))$

Not a delta function (coherent states are not orthogonal)

Overlap of two coherent states

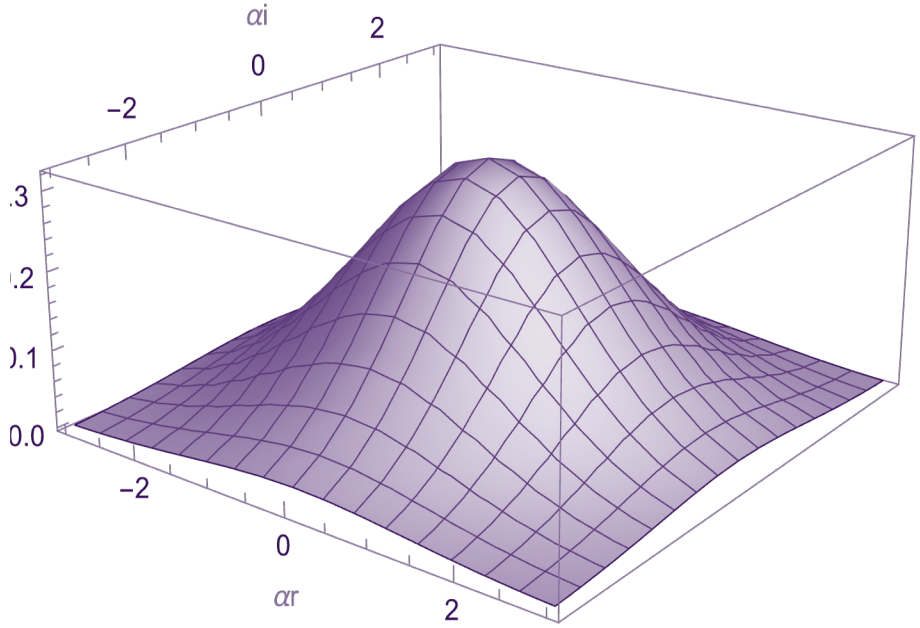
$$\langle\beta|\alpha\rangle = e^{-1\frac{1}{2}\left(|\beta|^2+|\alpha|^2-2\beta^*\alpha\right)} \neq \delta^2(\alpha-\beta)$$



Q function

Example: thermal state

$$\hat{\rho} = \frac{1}{\bar{n}+1} \sum_n \left(\frac{\bar{n}}{\bar{n}+1} \right)^n |n\rangle \langle n|$$



$$Q(\alpha) = \frac{1}{\pi} \langle \alpha | \hat{\rho} | \alpha \rangle$$

$$Q(\alpha) = \frac{1}{\pi} \frac{1}{\bar{n}+1} \sum_n \left(\frac{\bar{n}}{\bar{n}+1} \right)^n \langle \alpha | n \rangle \langle n | \alpha \rangle$$

Overlap of coherent state with number state

$$\langle n | \alpha \rangle = e^{-|\alpha|^2/2} \frac{\alpha^n}{\sqrt{n!}}$$

$$Q(\alpha) = \frac{1}{\pi} e^{-|\alpha|^2} \frac{1}{\bar{n}+1} \underbrace{\sum_n \left(\frac{\bar{n}}{\bar{n}+1} \right)^n \frac{|\alpha|^{2n}}{n!}}_{e^{\bar{n}|\alpha|^2/(1+\bar{n})}}$$

$$Q(\alpha) = \frac{1}{\pi} \frac{1}{\bar{n}+1} e^{-|\alpha|^2/(1+\bar{n})}$$

Broader gaussian centred at (0,0)

Q function

Example: Fock photon number state

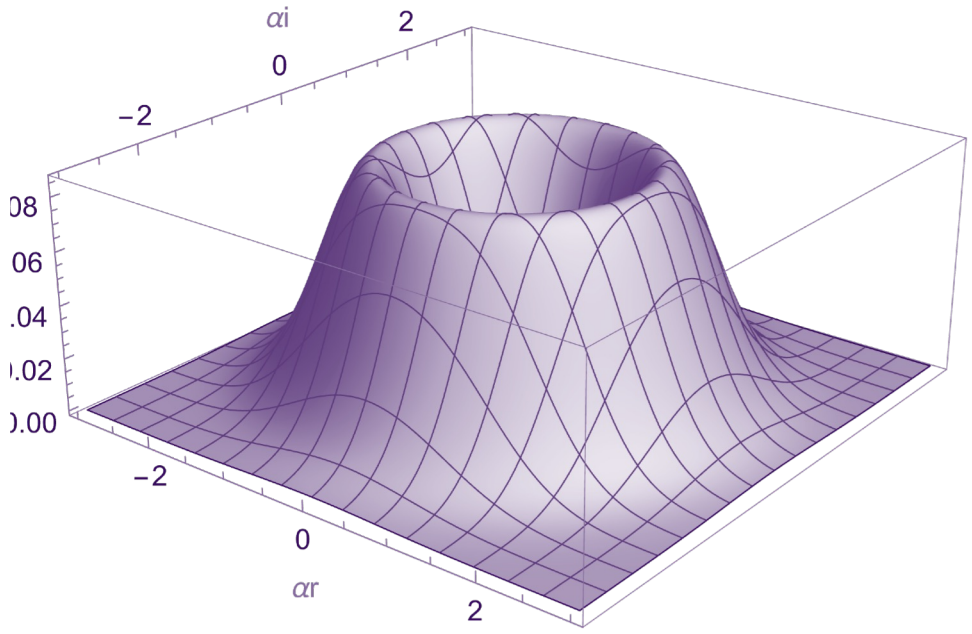
Q function of $|n\rangle$

$$Q(\alpha) = \frac{1}{\pi} |\langle n|\alpha\rangle|^2$$

$$\langle n|\alpha\rangle = e^{-|\alpha|^2/2} \frac{\alpha^n}{\sqrt{n!}}$$

$$Q(\alpha) = \frac{1}{\pi} e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}$$

Ring in phase space



Density matrix in the photon number basis

$$\hat{\rho} = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$$

p_i: probability of being in state $|\psi_i\rangle$

We can represent $\hat{\rho}$ in the photon number basis:

$$\hat{\rho} = \sum_n \sum_m \left|m\right\rangle \rho_{mn} \left\langle n\right|$$

$$\rho_{mn} = \left\langle m\right| \hat{\rho} \left|n\right\rangle$$

Diagonal elements are probabilities of finding n photons in the field

Glauber Sudarshan P function

Definition

We can represent $\hat{\rho}$ using coherent states

$$\hat{\rho} = \int P(\alpha) \left|\alpha\right\rangle \left\langle\alpha\right| \mathrm{d}^2\alpha$$

This is known as the P function

$$\mathrm{d}^2\alpha = \mathrm{d}\,\mathrm{Re}(\alpha)\,\mathrm{d}\,\mathrm{Im}(\alpha)$$

$\left|\alpha\right\rangle \left\langle\alpha\right|$ is a projection onto non-orthogonal states

$P(\alpha)$ is not a genuine probability distribution

There are states where $P(\alpha)$ is negative or highly singular, these states are non-classical

P function

Examples

$$\hat{\rho} = \int P(\alpha) \left|\alpha\right\rangle \left\langle\alpha\right| \mathrm{d}^2\alpha$$

Coherent state

$$\hat{\rho} = \left|\beta\right\rangle \left\langle\beta\right|$$

$$P(\alpha) = \delta^2 \left(\alpha - \beta\right)$$

Delta function

i.e. a single point in phase space

Photon number state

$$\hat{\rho} = \left|n\right\rangle \left\langle n\right|$$

$$P(\alpha) = \frac{e^{|\alpha|^2}}{n!} \frac{\partial^{2n}}{\partial\alpha^n \partial\alpha^{*n}} \delta^2(\alpha)$$

Derivative of a delta function

Sec. 3.7, of C. C. Gerry and Peter Knight Introductory quantum optics (2004)

P function

Q function can be obtained via convolution of the P function with a Gaussian:

$$Q(\alpha) = \frac{1}{\pi} \int d^2\beta \, P(\beta) e^{-|\alpha-\beta|^2},$$

Wigner function can be obtained via convolution of the P function with a Gaussian with a different width:

$$W(\alpha) = \frac{2}{\pi} \int d^2\beta \, P(\beta) e^{-2|\alpha-\beta|^2},$$

Wigner function

Definition

$W(x,p)$ behaves like a joint probability distribution

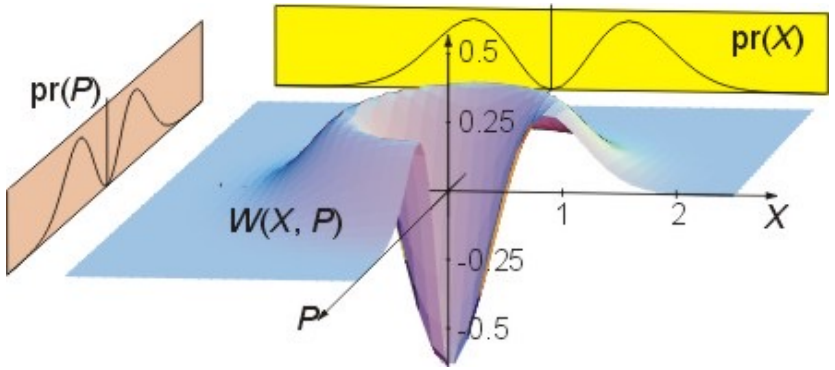
Can integrate over x or p to recover marginal distributions

$$P_x(x) = \int dp\, W(x,p); \quad P_p(p) = \int dx\, W(x,p),$$

$$\langle \hat{x}^s \rangle = \int dx\, x^s P_x(x); \quad \langle \hat{p}^s \rangle = \int dp\, p^s P_p(p),$$

The Wigner function can be negative

This indicates the state is non-classical



Wigner function of |1⟩

Wigner function

Examples

Coherent state

$$W(\alpha) = \frac{2}{\pi} e^{-2|\alpha-\beta|^2}$$

Gaussian centred at $(Re(\beta), Im(\beta))$

Gaussian is narrower than coherent state Q function

Thermal state

$$W(\alpha) = \frac{2}{\pi} e^{-2|\alpha|^2/(2\bar{n}+1)}$$

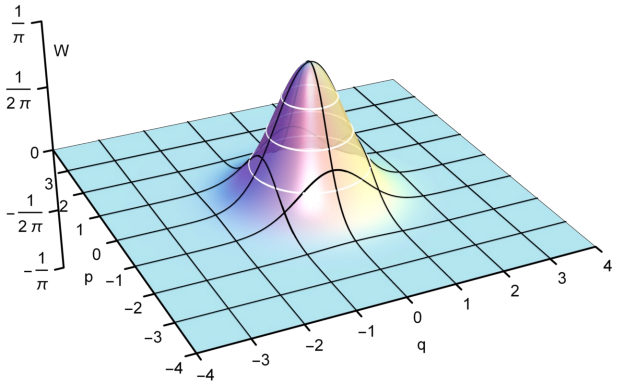
Photon number state

$$W(\alpha) = \frac{2(-1)^n}{\pi} e^{-2|\alpha|^2} L_n(4|\alpha|^2),$$

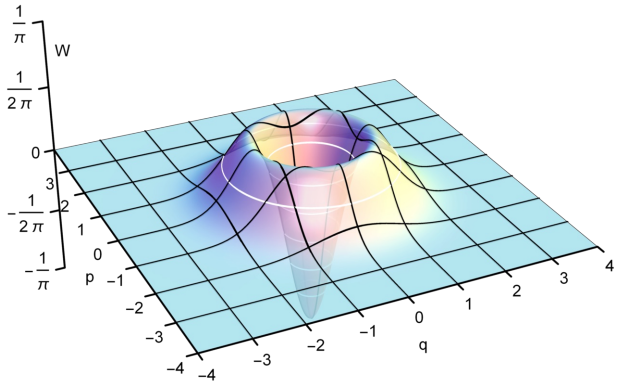
L_n is the Laguerre polynomial.

Wigner function

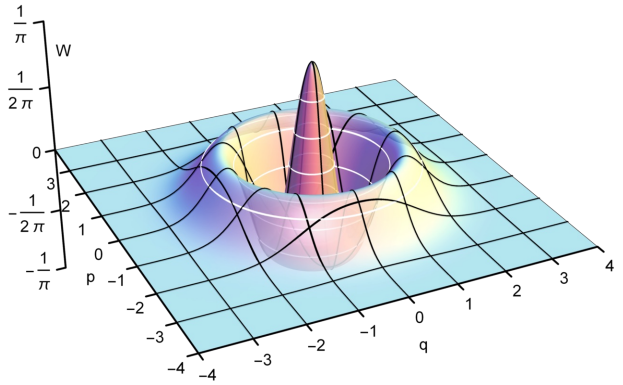
Fock states



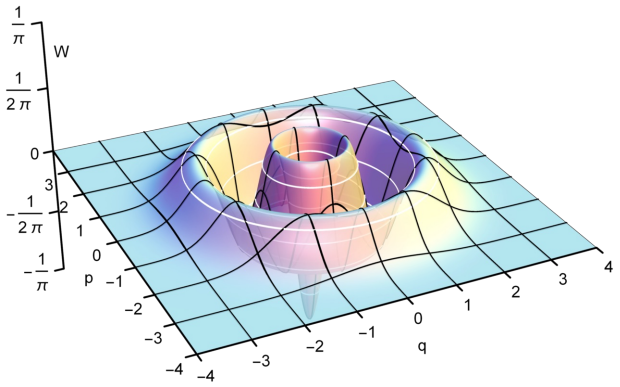
Wigner function of $|0\rangle$



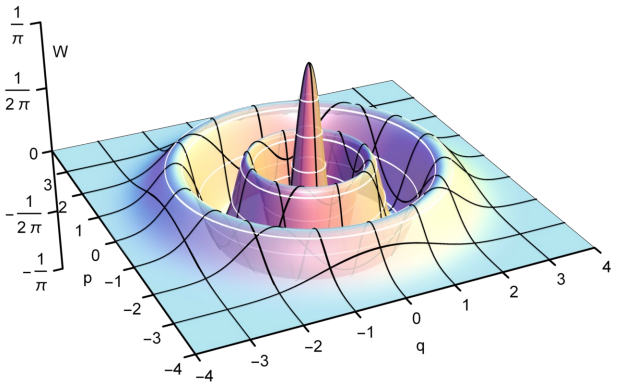
$|1\rangle$ (can see Wigner negativity)



$|2\rangle$



$|3\rangle$



$|4\rangle$

Comparing phase space representations

Coherent states

P function

$$P(\alpha) = \delta^2(\alpha - \beta)$$

Wigner

$$W(\alpha) = \frac{2}{\pi} e^{-2|\alpha-\beta|^2}$$

Q function

$$Q(\alpha) = \frac{1}{\pi} e^{-|\alpha-\beta|^2}$$

Wigner function is broader than P function

Q function is broader than Wigner function

Summary

In this lecture, we covered three different quasi-probability distributions used in quantum optics:

Husimi Q function

Q function can be thought of as a probability density
(P and W functions cannot)

Glauber Sudarshan P function

P function can be negative and singular

Wigner function

W function can be negative

Wigner function negativity is a sign of non-classicality eg. $|1\rangle$

